

ANDREW DEAL

Software Engineer + Technical Generalist

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Software engineer and technical generalist building AI systems that have to work outside the demo. Experience spans autonomous machinery, identity systems at consumer scale, edge ML, predictive maintenance, and a solo-built health-intelligence platform. Current focus: physical-AI deployment, evaluation, robot data, observability, and reliable operator tools.

BUILD TypeScript, JavaScript, Python, React, Node.js, FastAPI, SQL
OPERATE AWS, Docker, APIs, data pipelines, CI/CD, observability
INTEGRATE LLM orchestration, model evaluation, UX research, product strategy

EXPERIENCE

Independent Engineer

Physical AI · Self-employed
MAY 2026 TO PRESENT · MOUNTAIN VIEW, CA

- Trained a **52-million-parameter ACT policy** on 206 PushT episodes and 25,650 frames, preserving reproducible checkpoints and evaluation state.
- Built a fixed-seed rollout evaluation with videos, metrics, and explicit failure analysis. The initial ten-run result produced zero full task completions and partial geometric improvement in five runs.
- Brought up a SO-ARM101 leader and follower system through stable device routing, camera integration, and read-only communication with all six follower servos.
- Developing an inspectable deployment and evaluation artifact around multimodal data, failure classification, intervention, recovery, and operator acceptance.

Baseline Health

Founder and Product Engineer
FEBRUARY 2026 TO MAY 2026

- Built an AI-native health-intelligence system combining wearable data, laboratory results, and daily behavior into longitudinal scoring and coaching.
- Designed the full stack across React, TypeScript, Python, FastAPI, data ingestion, LLM orchestration, and deterministic audit paths, then onboarded founding members.
- Created a research layer separating analytical findings from member-facing claims for consequential recommendations.

Selected Hardtech

Pixel Foundry · OmenAI · TurbineOne
JANUARY 2024 TO JANUARY 2026

- Advised and led early-stage work across defense, construction, mining, AgTech, sensor-driven predictive maintenance, and edge ML in contested environments.
- Translated operator needs, sensor data, and operational constraints into product strategy, diagnostic surfaces, model priorities, and customer workflows.

Bear Flag Robotics

Product and Technical Lead · Senior Software Engineer
Acquired by John Deere
FEBRUARY 2020 TO MARCH 2023

- Co-led the web platform supporting **more than 80,000 autonomously tilled acres** and **more than \$1 million in gross revenue** across major enterprise farms.
- Designed full-stack systems and product workflows for autonomous field operations, customer visibility, and day and night deployment windows.
- Synthesized customer evidence and coordinated autonomy, tractor engineering, and field operations across Bear Flag and John Deere.

Over

Senior Software Engineer · Acquired by GoDaddy
MAY 2019 TO OCTOBER 2019

- Led web delivery for Over Teams, helped bring the enterprise product to parity with iOS, and contributed to threefold enterprise-account growth.
- Established frontend architecture and engineering standards for a design-led, cross-platform product.

Twitch

Software Engineer, Identity
MAY 2016 TO OCTOBER 2017

- Built OpenID Connect federation, username changes, registration experiments, and authentication-latency improvements in multi-team rollouts affecting **more than 10 million daily active users**.
- Led cross-team rollout work where product, trust, and reliability decisions had consumer-scale consequences.

The RSR Company

Technical Program Lead
JUNE 2013 TO JUNE 2014

- Led a global intranet rollout for Millennium Challenge Corporation and expansion of a Peace Corps partnership.

George Mason University
BA, Philosophy, 2012

University of Oxford
Distinguished Exchange Student Program, 2010–2011
Modern Philosophy, Logic, and Logic Systems

Reforge
Product Strategy and Product-Led Growth, 2022